# Poster: SurakshaNet: Privacy-First, Edge-Deployed Multilingual Abusive Language Detection

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## Abstract

Traditional abusive language moderation often relies on cloud-based APIs, raising privacy concerns and introducing network latency. We present *SurakshaNet*, an edge-deployed system for real-time multilingual toxicity detection on web messaging platforms, with a prototype implemented for WhatsApp Web. By combining INT8 static quantization with ONNX Runtime Web and WebAssembly (WASM), we reduce a 471 MB transformer model to 113 MB (about 76%) while incurring less than 0.11% accuracy loss. Implemented natively as a Chrome Extension using a Manifest V3 Service Worker, *SurakshaNet* achieves a median inference latency of 28 ms and ensures that raw user data never leaves the device.

## CCS Concepts

• **Security and privacy → Software and application security**.

## Keywords

Abusive language detection, Privacy-preserving AI, Edge computing

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## 1 Introduction

Malicious actors increasingly exploit social media platforms to spread harmful content [3], creating toxic online spaces where women, children, and other marginalized groups are frequently targeted. As gender-based harassment grows in code-mixed Indian contexts (Hinglish/Hindi) [1], accurate and timely abuse detection becomes essential for safe user interaction. There is an urgent need for proactive, privacy-preserving moderation tools. To address this, we develop *SurakshaNet*, a browser-based extension for real-time detection of harmful and abusive conversations.

Our contributions are:

- A lightweight web extension for in-browser abuse detection on web apps.
- A multilingual abuse detection pipeline with a size-optimized, edge-deployable model and forensic evidence support.

## 2 Implementation

The core intelligence follows a 5-step pipeline, as shown in Figure 1, beginning with multilingual abusive-language datasets and proceeding through model training. We trained our model on a combined dataset of 42,487 records, consisting of the MACD Hindi subset [1] and the Davidson English (EN) dataset (90% training split)[1]. The MACD dataset is a large-scale, human-annotated multilingual corpus from ShareChat, balanced with 49% abusive content. This combined training ensures the model is robust to both Hindi and English offensive language. A fine-tuned `MiniLM-L12-v2` is aggressively compressed using ONNX quantization [2], shrinking its RAM footprint to approximately 89 MB. This substantial reduction enables efficient browser-native execution through `onnxruntime-web`, making real-time edge inference practical. The model runs in a browser Service Worker, allowing for zero-network-dependency classification. A `MutationObserver` script monitors platforms like WhatsApp Web by injecting warning banners upon detecting toxic nodes. The optimized model is deployed as a Chrome extension, where ONNX-based in-browser inference supports fast, lightweight, and privacy-preserving detection of harmful content. This design makes the framework suitable for edge environments by avoiding cloud inference, reducing latency, and protecting sensitive user communication. Finally, the framework supports forensic evidence collection through DOM snapshots, chain-of-custody logs, AES-256 local encryption, and court-admissible incident report generation.

As shown in Figure 2, the system identifies both "Harmful" and "Critical" threats in multilingual contexts. It correctly classifies Hinglish/Hindi slang (e.g., catching abuse with 98.6% confidence) while maintaining a low false-positive rate for neutral conversational Hindi. The source code is available at Project Repository[2].

## 3 Empirical Evaluation

As shown in Table 1, the quantized model achieves a median latency of 28 ms and a compact memory footprint of approximately 89 MB, making it suitable for resource-constrained real-time inference. Our model achieves an accuracy of 84.66% on Hindi-only content (Test A) and 88.01% on Hindi+English content (Test B). The MACD paper reports 86.34% accuracy using a fine-tuned XLM-R Base model on Hindi-only data. While our edge model shows a 1.68% performance trade-off on Hindi-only tasks relative to this

[1] https://github.com/t-davidson/hate-speech-and-offensive-language
[2] https://github.com/PrathamGarg1/surkashanet

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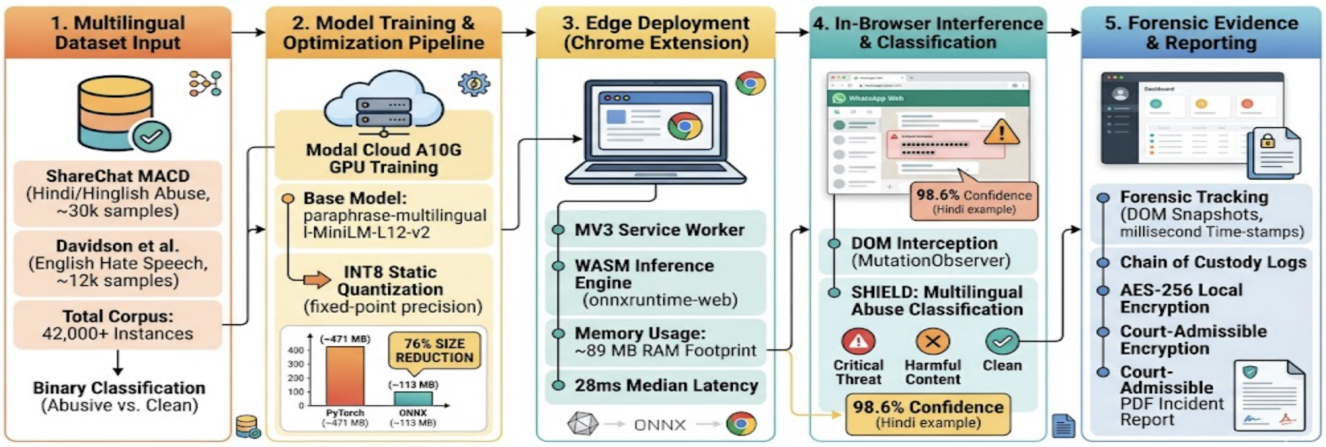



**Figure 1: End-to-end architecture of *SurakshaNet*, including multilingual dataset preparation, transformer model training, INT8 quantization, Chrome-based edge deployment via ONNX Runtime Web and WASM, real-time abuse detection through DOM interception, and forensic evidence logging.**

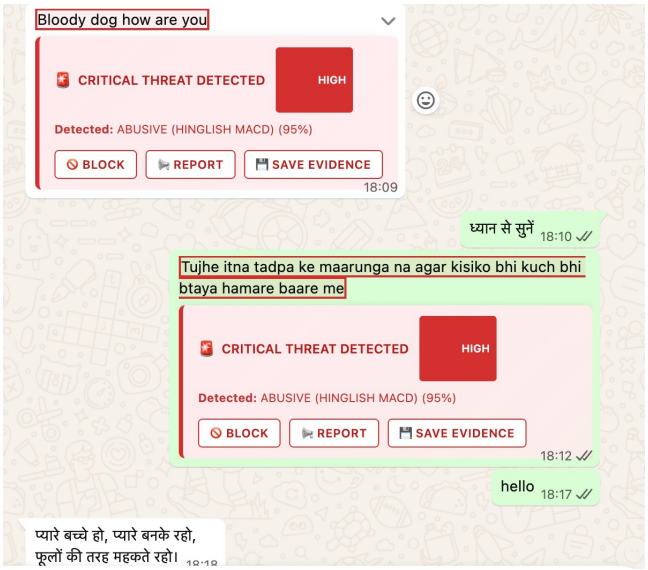



**Figure 2: Examples of *SurakshaNet* detecting abusive messages in real time on WhatsApp Web, showing severity-aware alerts.**

baseline, XLM-R Base (with approximately 270–278 million parameters) is infeasible for in-browser deployment due to its prohibitive memory footprint, far exceeding the RAM constraints of a browser runtime. Suitability for browser-native, privacy-first edge inference was the primary design constraint governing model selection. Crucially, for the multilingual Hindi+English setting, our model achieves 88.01%—surpassing the MACD baseline (which reports no English-inclusive accuracy)—demonstrating stronger generalization across languages. The observed accuracy degradation relative to the non-quantized version of our own model is negligible, remaining below 0.11%. In addition, the model size is reduced from 471 MB to 113 MB, corresponding to a 76% reduction, substantially improving storage efficiency and portability. All incidents are timestamped to millisecond precision and hashed (SHA-256) to ensure tamper-proofness. In the future, we will explore blockchain-based mechanisms to securely store forensic evidence in the cloud.

**Table 1: Post-Quantization Performance Summary**

| Metric | Value |
|---|---|
| Inference Latency | 28 ms |
| RAM Footprint | ~89 MB |
| Accuracy – Original Paper (XLM-R, Hindi) | 86.34% (not browser-deployable) |
| Accuracy (Test A: SurakshaNet, Hindi-only) | 84.66% |
| Accuracy (Test B: SurakshaNet, Hindi+Eng) | 88.01% (no baseline from paper) |
| Performance Delta (Hindi-only vs. [1]) | −1.68% |
| Quantization Loss | <0.11% (vs. non-quantized model) |
| Size Reduction | 76% (471 MB → 113 MB) |

## 4 Conclusion

SurakshaNet demonstrates that proactive, multilingual moderation is technically viable on the edge. By combining low-latency inference with forensic integrity, the system provides a comprehensive template for privacy-first safety tools in the browser.

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